

Networks

Computer representation

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Optimization: principles and algorithms



Computer representation

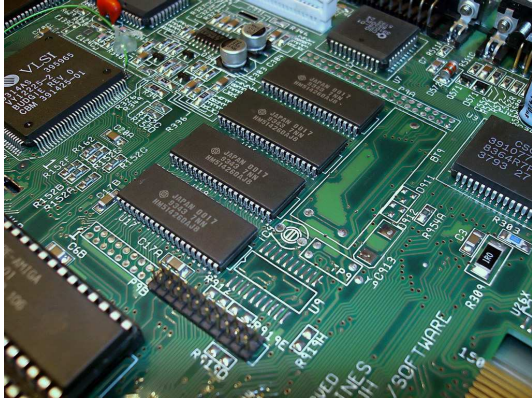
We usually use maps or schematics to look at networks

We obtain an overview of the overall topology.

But computers do not have this bird eyes's view

We present here two possible representations of networks in a computer.

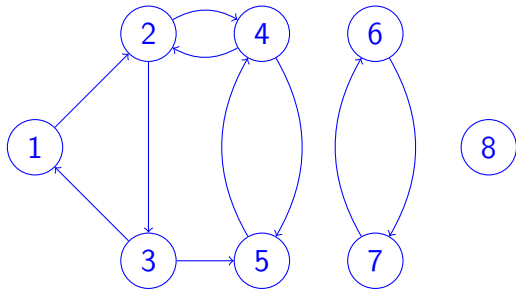
Computer representation



Adjacency matrix

$$A(i,j) = \begin{cases} 1 & \text{if } (i,j) \in \mathcal{A}, \\ 0 & \text{otherwise.} \end{cases}$$

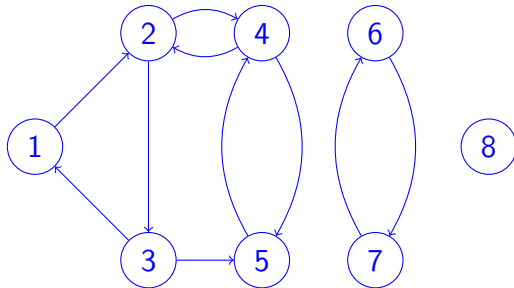
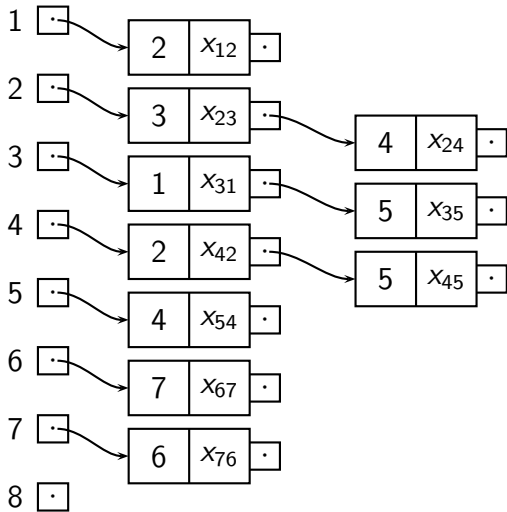
$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$



Adjacency matrix

- ▶ Valid representation because the incidence function is injective.
- ▶ Arc numbering convention for storing arc quantities.
- ▶ Sparse matrix: efficient storage techniques should be used.
- ▶ For instance, adjacency lists.

Adjacency lists



Summary

*Computers do not have a bird eye view of a network
Instead, they store the adjacency information using appropriate data structures.*